

Validity Analysis: INJONGO

Target context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Overall risk: **HIGH**

Deployment Context

Use case: Amharic utility chatbots for Ethiopian Electric Utility and Addis Ababa Water handling prepaid meter top-ups, outage reports, and bill disputes

Target population: Urban households in Addis Ababa and regional capitals interacting with utility support over Telegram and SMS

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology 	2	Significant gaps	high
Output Content 	2	Significant gaps	high
Output Form 	4	Minor gaps	high
Average	2.7		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

INJONGO is a serious, ground-up effort to build a culturally grounded African intent detection and slot-filling benchmark, and Amharic is one of its 17 supported languages. Dataset inspection confirms that the Amharic split contains genuine Ethiopian-grounded content: Telebirr appears as PAYMENT_COMPANY, Ethiopian banks (Amara, CBE) appear as BANK_NAME, Ethiopian regional cities and calendar months populate slot values, and Ethiopic script renders cleanly. However, for the specific deployment of an Amharic-language utility chatbot for EEU and AAWSA on Telegram/SMS, three high-priority dimensions (input ontology, input content, output ontology, output content) show substantive validity gaps. The 40-intent taxonomy lacks the deployment's primary use cases (prepaid meter top-up, STS token entry, outage report,

load-shedding schedule). The 23-slot ontology lacks all deployment-critical slot types (METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE). Code-switching — a confirmed high-priority deployment requirement — is absent from sampled Amharic data. Annotation was performed by pan-African crowdworkers without documented utility domain expertise, against a deployment requirement that ground truth be validated by EEU/AAWSA agents. Strongest dimensions are input form (text Ethiopic Unicode matches Telegram exactly) and output form (text intent+slots, standard metrics). The benchmark is best used as a fine-tuning starting point for general Amharic intent detection rather than a deployment-readiness evaluation for utility-specific tasks.

Practical Guidance

What This Benchmark Measures

INJONGO measures general-purpose Amharic intent detection (across banking, home, travel, kitchen & dining, and a generic utility domain) and slot-filling for 23 broad named-entity types. For the Ethiopian utility deployment, it can evaluate baseline Amharic NLU competence — whether a model recognizes Ethiopian banks, Telebirr, Ethiopian cities/calendar, formal polite Amharic register, and electricity/water bill_balance queries. Strongest dimensions are input form (Ethiopic Unicode in text matches Telegram deployment exactly) and output form (standard intent accuracy + slot F1 metrics).

Construct Depth

Construct depth is shallow for the deployment task. The benchmark probes general conversational NLU with cultural grounding but does not probe the specific constructs the deployment requires: prepaid meter workflows, STS token recognition, outage reporting, load-shedding queries, kebele-level address resolution, or Amharic-English code-switched technical vocabulary. The benchmark's 80-per-intent uniform distribution does not reflect real EEU/AAWSA traffic skew, so even on the intents that exist, the evaluation does not measure performance under deployment-realistic frequencies. Annotation is by general native speakers, not utility domain experts, so slot-boundary correctness for utility billing terminology is not validated against the authoritative ground-truth standard.

What Else You Need

Substantial supplementation is required. (1) Augment intent taxonomy with deployment-specific intents (input_ontology gap) using EEU/AAWSA conversation logs or expert-elicited examples. (2) Augment slot ontology with METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE, WOREDA, sub-city types (output_ontology gap). (3) Collect or synthesize Amharic-English code-switched utterances for utility domain technical vocabulary (input_content gap), since INJONGO's monolingual Amharic data will overestimate deployment performance. (4) Re-annotate or validate utility-specific slot boundaries with EEU/AAWSA customer service agents (output_content gap). (5) Build a deployment-distribution test set reflecting real intent frequencies, not uniform 80-per-intent. (6) Verify Unicode normalization (NFC/NFD) compatibility between INJONGO release files and the production tokenizer pipeline.

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

INJONGO includes a Utility domain explicitly selected as 'most suitable to African contexts' [Q39, Q117] and Amharic data does contain utility-adjacent bill_balance examples [D1, D8]. However, for the Ethiopian deployment specifically, the dimension is HIGH priority and the gaps are concrete and severe: prepaid meter top-up, STS token entry, outage reporting, load-shedding schedule queries, and account/meter lookup intents are not represented in the 19-example Amharic sample (D20) and are not enumerated in the documented intent list. The paper itself acknowledges domains 'essential for real-world applications' are missing [Q107] and notes the fixed uniform distribution may not reflect real-world frequencies [Q111]. Additionally, the bill_balance intent conflates utility billing with cable TV, retail credit, and phone bills across language splits (D1 vs D22), introducing construct-irrelevant variance for an EEU/AAWSA-scoped deployment. The 40-intent ceiling is also flagged as a source of measurement noise [Q86].

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g. " (p.3)
Q86	"For intent detection, we find that all open models achieved below 50% on the relatively easy task of intent detection. The poor performance may be attributed to either a lack of exposure to many African languages or the large label space (i.e. 40) for the classification task." (p.7)
Q107	"The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications." (p.9)
Q111	"Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations." (p.9)
Q117	"These are a total of 40 categories across 5 domains (Banking, Kitchen and Dining, Travel, Utility, and Home)." (p.13)
D1	"በወር ለመብራት ስንት ነው መክፈል የሚገባኝ?" — BILL_TYPE: ለመብራት — "How much do I need to pay for electricity per month?" — electricity utility bill query in Amharic
D8	"የዚህ ወር መክፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ።" — DATE: የዚህ ወር — "Show me the list of utility expenses to be paid this month." — direct utility bill balance query
D20	"ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንገረኝ።" — COUNTRY: ቱርክ — "Tell me a flight option to Turkey quickly." — Travel domain; no utility-specific intents observed in 19 Amharic samples
D22	"Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma" — RESTAURANT_NAME: taco bell — Hausa: "Can you confirm I have a table reservation at Taco Bell at 7 PM?" — Western fast food chain in culturally-specific dataset

Strengths

- Utility is one of five named domains [Q5, Q39] and the bill_balance intent contains genuine electricity utility content in Amharic (D1: ለመብራት annotated as BILL_TYPE)
- Cross-lingual evidence confirms electricity and water billing intents are operationally populated (D17 Luganda, D18 Kinyarwanda), so the utility category is not a stub
- Ground-up African design rather than translated CLINC reduces baseline Western bias in category selection [Q39, Q101]

ID	Evidence
Q5	"We cover the following five domains: banking, home, travel, utility, and kitchen & dining." (p.1)
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g." (p.3)
Q101	"Figure 4 shows our final experiments that compare cross-lingual transfer results from two English datasets: CLINC (Western-centric) and INJONGO (African-centric) on the intent detection task." (p.8)
D1	"በወር ለመብራት ስንት ነው መክፈል የሚገባኝ?" — BILL_TYPE: ለመብራት — "How much do I need to pay for electricity per month?" — electricity utility bill query in Amharic
D17	"Obukuumi n'amasannyalaze nteekeddwa kubisasula ddi?" — BILL_TYPE: Obukuumi, amasannyalaze — Luganda: "When do I need to pay the water and electricity bills?" — two BILL_TYPE slots; "amasannyalaze" = electricity
D18	"Mu bisanzwe nakwishyura fagitire y'amafaranga angahe ku mazi n'umuriro" — BILL_TYPE: mazi n'umuriro — Kinyarwanda: "How much do I normally pay for water and electricity bills?" — utility billing domain confirmed

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include prepaid meter top-up, STS token entry, outage report, outage status query, load-shedding schedule, bill dispute, payment history, account/meter lookup, payment via Telebirr/CBE Birr, water complaints, and human-agent escalation (per elicitation and YAML deployment_required_intents).

IO-2: Yes — confirmed omissions: no prepaid meter top-up, STS token, outage report, or load-shedding intents observed in 19 Amharic examples (D20) and not enumerated in documented intent list; paper explicitly notes 'healthcare and education' missing and 5/40 may miss real-world domains [Q107]. Flagged gap_id 1 in YAML resolved as NOT FOUND for utility-specific Ethiopian intents.

IO-3: Yes, partially — domains like Kitchen & Dining, Home (alarm, calendar_update, music), Travel (book_flight: D5, D20), and non-utility billing (cable TV, retail credit cards in bill_balance D22, D21) are present in the 40-intent space but irrelevant to an EEU/AAWSA chatbot, consuming evaluation weight when averaged over the 40-intent label space [Q86].

IO-4: Content underrepresentation gap: deployment-critical utility intents absent. Construct-irrelevant variance gap: heterogeneous bill_balance scope and non-utility domains inflate label space without serving deployment. Distributional mismatch: uniform 80-per-intent distribution [Q56] does not reflect skewed real-world utility interaction frequencies [Q111].

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g." (p.3)
Q56	"Our final annotation resulted in 3,200 annotated utterances, with 80 utterances per intent for each of the 16 African languages." (p.5)
Q86	"For intent detection, we find that all open models achieved below 50% on the relatively easy task of intent detection. The poor performance may be attributed to either a lack of exposure to many African languages or the large label space (i.e. 40) for the classification task." (p.7)

Q107	"The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications." (p.9)
Q111	"Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations." (p.9)
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D20	"ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንገረኝ።" — COUNTRY: ቱርክ — "Tell me a flight option to Turkey quickly." — Travel domain; no utility-specific intents observed in 19 Amharic samples
D21	"Ke hloka bokae ho lefa karete ya Edgars" — spans: [] — Sesotho: "How much do I need to pay the Edgars card?" — Edgars is South African retail credit card; no slots annotated despite implied BILL_TYPE/SHOPPING_ITEM
D22	"Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma" — RESTAURANT_NAME: taco bell — Hausa: "Can you confirm I have a table reservation at Taco Bell at 7 PM?" — Western fast food chain in culturally-specific dataset
WEB-11	gap_id 4 — no Ethiopian utility-specific NLP slot extraction literature exists; gap is documentation-confirmed (https://arxiv.org/pdf/2502.09814)

Information Gaps

- Full enumeration of the eight Utility-domain intent labels in INJONGO is not surfaced (gap_id 1)
- Full Amharic intent distribution beyond 19 sampled examples is unknown — utility-domain content density at scale is unquantified

Requires Expert Verification

- EEU/AAWSA production intent frequency distribution to compare against INJONGO's uniform 80-per-intent design
- Whether INJONGO's documented utility intents (e.g., 'pay bill') map onto EEU prepaid top-up workflow conventions

Remediation

Gap 1: Deployment-critical utility intents (prepaid meter top-up, STS token entry, outage report, load-shedding schedule query, account/meter lookup) are absent from the documented 40-intent taxonomy.

Recommendation: Define an extended intent set with EEU and AAWSA customer service leads; collect 80–200 seed utterances per new intent in Amharic via supervised elicitation from Ethiopian native speakers familiar with the utility workflow; concatenate to INJONGO Amharic data for fine-tuning.

Gap 2: Uniform 80-per-intent distribution does not reflect real EEU/AAWSA traffic skew (likely heavily weighted toward top-up and outage queries).

Recommendation: Build a frequency-weighted evaluation set sampled in proportion to observed EEU/AAWSA chatbot or call-center intent distribution; report per-intent metrics alongside macro-averaged accuracy to surface deployment-relevant performance.

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Mixed evidence with one decisive negative for this HIGH-priority dimension. INJONGO was designed with cultural grounding as an explicit principle [Q13, Q30, Q37], and dataset inspection confirms Ethiopian-specific entities in the Amharic split: Telebirr as PAYMENT_COMPANY [D4, D11], Ethiopian banks Amara Bank and CBE [D2, D3], Ethiopian cities Dire Dawa, Bahir Dar, Hawassa [D5, D12], and Ethiopian calendar months Hamle and Meskerem [D5, D6] — these resolve key gaps from web search positively. However, the deployment-critical requirement of Amharic-English code-switching is not represented: zero of 19 Amharic samples contain Latin-script tokens or English insertions (D24), and code-switching was not an explicit design criterion in the paper. The deployment elicitation marks code-switching as 'common and increasing' particularly for technical vocabulary (token, top-up, balance, meter), and a benchmark monolingual in Ethiopic script will overestimate performance on naturalistic deployment input. Additionally, regional dialectal variation (Oromo/Tigrinya substrate) is not addressed, and Amharic is acknowledged as systematically lower-resource than Swahili [Q88, Q89].

ID	Evidence
Q13	"However, these efforts face two key challenges: (1) the translationese effect, which makes utterances sound less natural in the target languages (Vanmassenhove et al., 2021; Bizzoni et al., 2020), and (2) the creation of utterances that are less culturally relevant." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
Q37	"To simplify the process while making the dataset cultural, we provide each annotator with sample sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, say "transfer"." (p.3)
Q88	"The performance of some African languages is often higher than others, this is probably connected to the amount of monolingual data available on the web." (p.7)
Q89	"For example, Swahili (swa) with over 1 billion monolingual data (Kudugunta et al., 2023) has 80.6 accuracy point that is comparable performance to English performance (81.1) with Llama 3.3 70B, while other languages have much lower performance." (p.7)
D2	"የአማራ ባንክ አካውንቱ ስንት አለው" — BANK_NAME: አማራ ባንክ — "How much is in my Amara Bank account?" — Ethiopian bank (Amara Bank) named as BANK_NAME slot
D3	"ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ" — BANK_NAME: ንግድ ባንክ — "Is there money remaining at Commercial Bank?" — CBE (Commercial Bank of Ethiopia) named
D4	"በቴሌብር የብድር አካውንቱ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY
D5	"ከድረዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድረዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D6	"ከሃምሌ 4 እስከ 6 የቤት ኪራይ ማግኘት እችላለሁ?" — DATE: ከሃምሌ 4 እስከ 6 — "Can I get a home car rental from Hamle 4 to 6?" — Ethiopian calendar month (Hamle) used for DATE slot
D11	raw: "ከቴሌ ብር አካውንቱ ገንዘብ ለማን ለማን እንደላኩ ዝርዝሩን አሳየኝ?" text: "Show me history of my telebirr transfers?" — PAYMENT_COMPANY: telebirr — Telebirr confirmed in Amharic source utterance as code-switched reference; PAYMENT_COMPANY slot correctly labeled
D12	raw: "ለቀጣይ የጉብኝት ጉዞ በሃዋሳ እና በአርባ ምንጭ መካከል እንድመርጥ..." text: "help me decide between Hawassa and Arbaminch..." — CITY_OR_PROVINCE: Hawassa, Arbaminch — Ethiopian regional cities (Hawassa, Arba Minch) correctly identified in travel context
D24	All 19 Amharic examples use Ethiopic script without any romanized characters or Latin-script code-switching — No code-switched (Amharic + English intrasentential) utterances observed in 19 Amharic train examples

Strengths

- Explicit cultural-grounding methodology, with native-speaker generation rather than translation [Q37, Q40, Q41]
- Telebirr confirmed as PAYMENT_COMPANY value in Amharic data (D4) — a high-priority deployment entity
- Ethiopian banks (Amara Bank, CBE) and Ethiopian regional cities (Dire Dawa, Bahir Dar, Hawassa) appear as slot values [D2, D3, D5, D12]
- Ethiopian calendar months (Hamle, Meskerem) appear as DATE values [D5, D6], reflecting authentic Amharic temporal expression
- Formal polite Amharic register present in samples (D29), matching deployment norms for formal service interaction

ID	Evidence
Q37	"To simplify the process while making the dataset cultural, we provide each annotator with sample sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, say "transfer". " (p.3)
Q40	"A Xhosa annotator was asked to generate another utterance belonging to the same intent type but capturing the South African context where the language is spoken." (p.3)
Q41	"Thus, the annotator used the R200 as "money" with currency Rand (R), and more familiar South African banks such as "FNB" and "Absa" for "bank name" slot." (p.3)
D2	"የአማራ ባንክ አካውንቲ ስንት አለው" — BANK_NAME: አማራ ባንክ — "How much is in my Amara Bank account?" — Ethiopian bank (Amara Bank) named as BANK_NAME slot
D3	"ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ" — BANK_NAME: ንግድ ባንክ — "Is there money remaining at Commercial Bank?" — CBE (Commercial Bank of Ethiopia) named
D4	"በቴሌብር የብድር አካውንቲ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D6	"ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ።" — DATE: ከሃምሌ 4 እስከ 6 — "Can I get a home car rental from Hamle 4 to 6?" — Ethiopian calendar month (Hamle) used for DATE slot
D12	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል እንድመርጥ..." text: "help me decide between Hawassa and Arbaminch..." — CITY_OR_PROVINCE: Hawassa, Arbaminch — Ethiopian regional cities (Hawassa, Arba Minch) correctly identified in travel context
D29	"የእኔ የወጭ አካውንት ቀሪ ሂሳብ ስንት እንደሆነ ሊነግሩኝ ይችላሉ።" — ACCOUNT_TYPE: የወጭ አካውንት — "Can you tell me the remaining balance of my expense account?" — formal polite register (ሊነግሩኝ ይችላሉ)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — deployment requires Ethiopian-specific cultural/geographic knowledge (Ethiopian payment platforms, banks, kebele/sub-city addresses, Ethiopian calendar) and dialectal robustness (Amharic-English code-switching, possible Oromo/Tigrinya substrate).

IC-2: Partially — Ethiopian financial entities (Telebirr D4, Amara Bank D2, CBE D3) and geography [D5, D12] are present, satisfying baseline cultural grounding [Q30]. However, code-switching is absent (D24) and regional dialectal variation is undocumented (gap_id 6 NOT FOUND).

IC-3: Some Western residue persists (e.g., 'taco bell' in Hausa D22), though for the Amharic split the sampled instances appear consistently Ethiopia-grounded. Cross-lingual training data drawn from 'Combined INJONGO' [Q72, Q128] could expose models to non-Ethiopian entities.

IC-4: INSUFFICIENT DOCUMENTATION — paper documents pan-African native-speaker recruitment via Kenya-based logistics [Q44]; whether Amharic annotators were Ethiopia-based or had urban Addis Ababa familiarity is not specified. Would need annotator location/demographic breakdown.

IC-5: Major content issue: zero observed code-switching across 19 Amharic examples (D24) despite deployment confirming high code-switching frequency for English domain vocabulary. This represents a content distribution mismatch likely to harm external validity. Secondary issue: Amharic web-data resource asymmetry vs. higher-resource African languages may compound performance gaps [Q88, Q89].

ID	Evidence
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
Q41	"Thus, the annotator used the R200 as "money" with currency Rand (R), and more familiar South African banks such as "FNB" and "Absa" for "bank name" slot." (p.3)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q72	"The dataset of SFT was obtained by aggregating all the training samples of the 17 languages in INJONGO i.e. "Combined INJONGO", with randomly sampled prompts from a pool of 5." (p.6)
Q88	"The performance of some African languages is often higher than others, this is probably connected to the amount of monolingual data available on the web." (p.7)
Q89	"For example, Swahili (swa) with over 1 billion monolingual data (Kudugunta et al., 2023) has 80.6 accuracy point that is comparable performance to English performance (81.1) with Llama 3.3 70B, while other languages have much lower performance." (p.7)
Q128	"Training data is constructed from the combined train splits of INJONGO dataset across all 17 languages, with prompts randomly sampled from a pool of 5 predefined templates." (p.14)
D2	"የአማራ ባንክ አካውንቲ ስንት አለው" — BANK_NAME: አማራ ባንክ — "How much is in my Amara Bank account?" — Ethiopian bank (Amara Bank) named as BANK_NAME slot
D3	"ንግድ ባንክ ላይ የቀረጃ ገንዘብ አለ" — BANK_NAME: ንግድ ባንክ — "Is there money remaining at Commercial Bank?" — CBE (Commercial Bank of Ethiopia) named
D4	"በቴሌብር የብድር አካውንቲ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D12	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል እንድመርጥ..." text: "help me decide between Hawassa and Arbaminch..." — CITY_OR_PROVINCE: Hawassa, Arbaminch — Ethiopian regional cities (Hawassa, Arba Minch) correctly identified in travel context
D22	"Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma" — RESTAURANT_NAME: taco bell — Hausa: "Can you confirm I have a table reservation at Taco Bell at 7 PM?" — Western fast food chain in culturally-specific dataset
D24	All 19 Amharic examples use Ethiopic script without any romanized characters or Latin-script code-switching — No code-switched (Amharic + English intrasentential) utterances observed in 19 Amharic train examples
WEB-5	Telebirr 54.84M users, dominant Ethiopian utility payment channel — confirms deployment relevance of Telebirr coverage (https://techafricanews.com/2025/07/25/ethio-telecom-achieves-83-2m-subscribers-and-73-revenue-surge-as-lead-strategy-concludes/)
WEB-9	INJONGO paper — gap_id 3 NOT FOUND on Amharic annotator domain expertise; gap_id 7 NOT FOUND on Amharic-specific Kappa (https://arxiv.org/abs/2502.09814)
WEB-12	Code-switching NLP literature — no Amharic-English Ethiopian utility code-switching benchmark exists (https://github.com/gentaiscool/code-switching-papers)

Information Gaps

- Frequency of Telebirr/CBE Birr appearances at full-corpus scale (sample shows present but not density)
- Regional dialectal variation in Amharic split (gap_id 6 NOT FOUND) — Addis Ababa vs Dire Dawa/Hawassa/Mekelle representation
- Whether any utility-specific Amharic content beyond bill_balance exists in unsampled portions of the 2,240 train examples
- Code-switching density across full Amharic split — sample of 19 may understate

Requires Expert Verification

- Whether sampled Amharic register and lexical choices reflect EEU/AAWSA customer service interactions specifically
 - Whether informal Addis Ababa neighborhood naming (Bole, Piassa, Megenagna) appears in PLACE_NAME slots in unsampled data
-

Remediation

Gap 1: Zero observed Amharic-English code-switched utterances in sampled INJONGO Amharic, despite code-switching being a confirmed high-priority deployment pattern especially for technical vocabulary (token, top-up, balance, meter).

Recommendation: Augment training data with code-switched Amharic-English variants, ideally elicited from younger urban Telegram users; ensure technical English domain vocabulary appears in mixed-script form; evaluate a held-out code-switched test split separately from monolingual.

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Strongest overall match. INJONGO is text-only [Q43] aligning with Telegram/SMS deployment, and dataset inspection confirms Ethiopic Unicode renders correctly without artifacts in all 19 sampled Amharic examples [D29, D5]. Annotation occurred on LabelStudio in native script [Q43]. Mixed Ethiopic/Arabic numeral usage is observed incidentally (D27), reflecting a pattern relevant to deployment without being formally designed. Two minor concerns moderate the score: (a) Unicode normalization form (NFC/NFD) is undocumented (gap_id 5 NOT FOUND), which could cause tokenizer mismatches in deployment, and (b) eng config exhibits transliteration degradation (D10: 'Dashen' → 'dasion'), which affects translate-test cross-lingual evaluation but not the primary in-language Amharic deployment path. SMS-specific signal degradation is not addressed in documentation, but per elicitation Telegram is the dominant primary channel and the deployment explicitly excludes romanized fallback (Q4 elicitation).

ID	Evidence
Q4	"We develop INJONGO—the first large-scale multicultural intent detection and slot-filling dataset covering 16 African languages, and English language." (p.1)
Q43	"The annotation followed the named entity recognition setup on LabelStudio platform." (p.4)
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D10	raw: "እባካችሁ የዳሽን ባንክ ሂሳቤን ታረጋግጡልኝላችሁ?" text: "can you please check my dasion bank account" — BANK_NAME: dasion bank — Original Amharic cites Dashen Bank; English translation renders "dasion" (spelling error) — shows transliteration degradation in eng config
D27	"ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ።" — DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car for 3 months starting next month." — Arabic numeral "3" mixed with Ethiopic in DATE span
D29	"የእኔ የወጭ አካውንት ቀሪ ሂሳብ ስንት እንደሆነ ሊነግሩኝ ይችላሉ።" — ACCOUNT_TYPE: የወጭ አካውንት — "Can you tell me the remaining balance of my expense account?" — formal polite register (ሊነግሩኝ ይችላሉ)

Strengths

- Text-only modality matches Telegram/SMS deployment exactly [Q43]
- Ethiopic script renders correctly in all 19 sampled Amharic examples without encoding artifacts [D29, D5]
- Native-script annotation on LabelStudio [Q43]
- Incidental coverage of mixed Ethiopic/Arabic numerals (D27) reflects realistic input pattern
- License (CC BY 4.0 [Q23]) permits adaptation for production fine-tuning

ID	Evidence
Q23	"Dataset is released under CC BY 4.0 license." (p.2)
Q43	"The annotation followed the named entity recognition setup on LabelStudio platform." (p.4)
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D27	"ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ።" — DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car for 3 months starting next month." — Arabic numeral "3" mixed with Ethiopic in DATE span

D29	"የእኔ የወጭ አካውንት ቀሪ ሂሳብ ስንት እንደሆነ ሊነግሩኝ ይችላሉ።" — ACCOUNT_TYPE: የወጭ አካውንት — "Can you tell me the remaining balance of my expense account?" — formal polite register (ሊነግሩኝ ይችላሉ)
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution match is good for primary channel: text in Ethiopic Unicode confirmed (D29). SMS-specific degradation (encoding loss, length truncation) is not represented in benchmark, but deployment scopes SMS as secondary and full Ethiopic as required [elicit¹ation Q4].

IF-2: Yes — Telegram and Ethio Telecom infrastructure support Ethiopic Unicode reliably; 4G coverage exceeds 80% in primary deployment cities [WEB-6, WEB-8]; Telegram transmits Ethiopic reliably (YAML).

IF-3: Numeral mixing observed (D27); Unicode normalization undocumented (gap_id 5); SMS Ethiopic transmission reliability may vary by carrier — not benchmarked. eng-config translate-test path shows transliteration noise (D10) that would affect cross-lingual transfer evaluation but not in-language deployment.

IF-4: Minor form gaps: (a) NFC/NFD normalization unspecified — could cause tokenizer subword mismatch; (b) no SMS-degraded variants tested; (c) romanized Amharic explicitly excluded from deployment scope so absence is acceptable.

ID	Evidence
Q4	"We develop INJONGO—the first large-scale multicultural intent detection and slot-filling dataset covering 16 African languages, and English language." (p.1)
Q43	"The annotation followed the named entity recognition setup on LabelStudio platform." (p.4)
D10	raw: "እባክሽሁ የዳሽን ባንክ ሂሳቤን ታረጋግጡልኛለሁ?" text: "can you please check my dasion bank account" — BANK_NAME: dasion bank — Original Amharic cites Dashen Bank; English translation renders "dasion" (spelling error) — shows transliteration degradation in eng config
D27	"ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ ወኪና ወኪራየት እፈልጋለሁ።" — DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car for 3 months starting next month." — Arabic numeral "3" mixed with Ethiopic in DATE span
D29	"የእኔ የወጭ አካውንት ቀሪ ሂሳብ ስንት እንደሆነ ሊነግሩኝ ይችላሉ።" — ACCOUNT_TYPE: የወጭ አካውንት — "Can you tell me the remaining balance of my expense account?" — formal polite register (ሊነግሩኝ ይችላሉ)
WEB-3	McGill-NLP/Injongo GitHub — Amharic example አባይ ባንክ confirms Ethiopic rendering at release (https://github.com/McGill-NLP/Injongo)
WEB-6	4G coverage >80% in primary deployment cities — text channel infrastructure available (https://www.mordorintelligence.com/industry-reports/ethiopia-telecom-mno-market)
WEB-8	Ethio Telecom FY2024/25 — 1,683 new mobile sites, 512 cities with 4G (https://addisinsight.net/2025/07/24/ethio-telecom-reports-162-billion-birr-revenue-and-4-9-trillion-birr-in-telebirr-transfers-in-2017-fiscal-year/)

Information Gaps

- Unicode normalization form (NFC/NFD) of release files (gap_id 5 NOT FOUND)
- Whether SMS channel Ethiopic delivery reliability has been measured by EEU/AAWSA
- Byte-offset vs character-offset behavior for span annotations in Ethiopic UTF-8 (dataset analysis Limitation 5)

Requires Expert Verification

- Whether tokenizers used in deployment match the byte-level normalization of INJONGO release files
-

Remediation

Gap 1: Unicode normalization form (NFC vs NFD) of INJONGO Amharic release is undocumented, which can cause tokenizer subword mismatch in production.

Recommendation: Audit the parquet files for Unicode normalization, normalize to a single form (NFC recommended for Ethiopic), and verify tokenizer behavior on Telegram-sourced production input matches the normalized training data.

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MODERATE-priority dimension with concrete structural gaps. The 23-slot ontology was rigorously developed with merge consolidation [\[Q54, Q55, Q119\]](#) and includes deployment-relevant types (BANK_NAME, PAYMENT_COMPANY, BILL_TYPE, MONEY, ACCOUNT_TYPE, CITY_OR_PROVINCE, PLACE_NAME, DATE, TIME). However, the dimension fails on critical deployment requirements: dataset inspection confirms (D19) that no METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE, WOREDA, or sub-city slot types exist in any of 19 Amharic examples, and these are core to EEU prepaid top-up and AAWSA billing workflows. Nine 'infrequent slots' were eliminated [\[Q52\]](#) without documented utility-deployment justification. Cross-lingual annotation choices show ontological ambiguity: SNEL (DRC electricity utility) is annotated as PAYMENT_COMPANY (D28), conflating utility provider with payment channel — relevant if EEU/AAWSA appear similarly. The deployment elicitation flags this as MODERATE priority because BANK_NAME/PAYMENT_COMPANY/BILL_TYPE provide partial coverage, but the missing types are non-negotiable for the core workflow.

ID	Evidence
Q52	"Consequently, nine infrequent slots from NATIONALITY through PLUG TYPE were eliminated." (p.4)
Q54	"Appendix A.2 contains all 34 slot types selected." (p.4)
Q55	"This merging process resulted in a reduction from 34 to 23 slot types." (p.5)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)
D19	በወር ለውጥ ስንት ነው መክፈል የሚገባኝ? — No METER_ID, UTILITY_ACCOUNT_NUMBER, or STS_TOKEN slots present anywhere in Amharic sample
D28	"Niongo ya snel ezali 2000 FC Yebisa ngai soki nakosi" — PAYMENT_COMPANY: snel, MONEY: 2000 FC — Lingala: "SNEL bill is 2000 FC, tell me if I've paid" — SNEL is the DRC national electricity utility; utility company annotated as PAYMENT_COMPANY

Strengths

- 23 slot types include deployment-relevant categories: BANK_NAME, PAYMENT_COMPANY, BILL_TYPE, MONEY, ACCOUNT_TYPE, CITY_OR_PROVINCE, PLACE_NAME, DATE, TIME, NUMBER [\[Q144\]](#)
- Telebirr fits PAYMENT_COMPANY (D4) and BILL_TYPE accommodates electricity (D1) — partial coverage of deployment slot space
- Annotation guide provides explicit disambiguation rules (e.g., temporal: <1 day → TIME; ≥1 day → DATE [\[Q153, Q154\]](#); bank name excludes 'bank' unless attached [\[Q152\]](#))
- Slot consolidation 34→23 was driven by frequency analysis [\[Q51, Q119\]](#), reducing label noise

ID	Evidence
Q51	"We decided to exclude slot entities appearing less than 500 times across all languages (after MUSIC_GENRE in the figure)." (p.4)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)

Q144	"# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19)
Q152	"When annotating Bank names, you do not need to include "bank" unless it is attached to the bank name, like seen above, with Ecobank." (p.21)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
D1	"በወር ለመብራት ስንት ነው መክፈል የሚገባኝ?" — BILL_TYPE: ለመብራት — "How much do I need to pay for electricity per month?" — electricity utility bill query in Amharic
D4	"በቴሌብር የብድር አካውንቱ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Deployment-relevant present categories: BANK_NAME, PAYMENT_COMPANY, BILL_TYPE, MONEY, ACCOUNT_TYPE, CITY_OR_PROVINCE, PLACE_NAME, DATE, TIME, NUMBER [Q144].

OO-2: Confirmed missing for Ethiopian utility deployment (D19): METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN (20-digit prepaid token), KEBELE, WOREDA, sub-city, NEIGHBORHOOD_NAME (informal Addis Ababa toponyms). The annotation guide requires explicit named entities for PLACE_NAME [Q156], so informal kebele/neighborhood references may not even fit existing types cleanly.

OO-3: Some category encodings reflect non-Ethiopian assumptions: PAYMENT_COMPANY conflated with utility provider (D28: SNEL); SHOPPING_ITEM, MUSIC_GENRE, ARTIST_NAME, DISH_OR_FOOD, MEAL_PERIOD relevant only to non-utility domains. Nine slots (NATIONALITY through PLUG TYPE) were eliminated [Q52] — without justification of trade-off against utility-specific slots.

OO-4: Stakeholder-driven taxonomy redesign is required. EEU customer service agents and AAWSA billing staff should define slot schemas for METER_ID format, account-number format, kebele granularity, and STS token recognition.

OO-5: Structural validity issue: deployment-critical decision boundaries (meter ID extraction, STS token recognition, kebele resolution) cannot be expressed in the 23-slot taxonomy, so benchmark performance does not reflect deployment task structure. Content validity issue: 9 missing slot types are deployment-critical.

ID	Evidence
Q52	"Consequently, nine infrequent slots from NATIONALITY through PLUG TYPE were eliminated." (p.4)
Q55	"This merging process resulted in a reduction from 34 to 23 slot types." (p.5)
Q144	"# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19)
Q156	"With this entity, only annotate if entity is named explicitly, e.g Name of airport, museum or mall is not and nt just "mall", "airport" etc" (p.24)

D19	በወር ለመብራት ስንት ነው መክፈል የሚገባኝ? — No METER_ID, UTILITY_ACCOUNT_NUMBER, or STS_TOKEN slots present anywhere in Amharic sample
D28	"Niongo ya snel ezali 2000 FC Yebisa ngai soki nakosi" — PAYMENT_COMPANY: snel, MONEY: 2000 FC — Lingala: "SNEL bill is 2000 FC, tell me if I've paid" — SNEL is the DRC national electricity utility; utility company annotated as PAYMENT_COMPANY

Information Gaps

- EEU and AAWSA's actual slot schemas for meter IDs, account numbers, and addresses (not in public sources)
- Whether any pre-merge slot types (among the dropped 9) would have been useful for utility deployment

Requires Expert Verification

- EEU billing system's canonical slot taxonomy and how informal Amharic place references should be normalized
- AAWSA's account-number and address conventions

Remediation

Gap 1: 23-slot ontology lacks METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, KEBELE, WOREDA, sub-city, and NEIGHBORHOOD_NAME types — confirmed absent in dataset (D19).

Recommendation: Co-design an extended slot schema with EEU and AAWSA billing staff; add at minimum METER_ID, UTILITY_ACCOUNT_NUMBER, STS_TOKEN, and KEBELE; re-annotate the existing Amharic split or annotate a deployment-specific augmentation under the extended schema.

ID	Evidence
D19	በወር ለመብራት ስንት ነው መክፈል የሚገባኝ? — No METER_ID, UTILITY_ACCOUNT_NUMBER, or STS_TOKEN slots present anywhere in Amharic sample

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

HIGH-priority dimension with substantial concerns. INJONGO followed a rigorous quality control process (three annotators per utterance [\[Q42, Q45\]](#), two review rounds [\[Q109\]](#), post-review Kappa 0.912–1.00 [\[Q48\]](#)). However, for the Ethiopian utility deployment specifically: (1) Ground-truth authority mismatch — annotators were recruited via a Kenya-based logistics company at per-country rates [\[Q44\]](#), and there is no documentation that Amharic annotators had EEU/AAWSA utility billing domain expertise (gap_id 3 NOT FOUND); the deployment elicitation explicitly requires utility agent validation. (2) Annotation quality issues observable in samples — D15 and D16 show Amharic utterances with clearly extractable DATE, TIME, and CALENDAR_EVENT entities but empty span annotations; D13 (Oromo) and D21 (Sesotho) show similar omissions. (3) Amharic-specific Fleiss Kappa not surfaced (gap_id 7 NOT FOUND), so where Amharic falls in the 0.618–0.934 pre-review range is unknown. (4) Pre-to-post improvements were largest in Sesotho and Zulu [\[Q49\]](#), suggesting language-specific quality variation that is not transparent for Amharic.

ID	Evidence
Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
Q49	"Notable improvements were observed in Sesotho (+0.327) and Zulu (+0.294), with other languages showing average improvements of approximately 0.1 in their Fleiss' Kappa scores." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
D13	"Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyaa irraa naaf kuti" — no spans — Oromo: "Book me a flight from Dire Dawa to Bole then to Jimma on Ethiopian Airlines" — Bole airport and Dire Dawa named, spans incorrectly empty
D15	"ነገ ጠዋት 12 ሰዓት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት" — spans: [] — "Increase the volume of the alarm I set for 12 o'clock tomorrow morning so I can hear it well." — TIME and DATE slots present in text but no spans annotated
D16	"የእናቴን እና የአባቴን የልደት ቀን የቀን መቁጠሪያ ላይ ስላቀያየርኩት አስተካክለዋል" — spans: [] — "Fix it because I changed my mother's and father's birthday on my calendar." — CALENDAR_EVENT-like content but no spans annotated
D21	"Ke hloka bokae ho lefa karete ya Edgars" — spans: [] — Sesotho: "How much do I need to pay the Edgars card?" — Edgars is South African retail credit card; no slots annotated despite implied BILL_TYPE/SHOPPING_ITEM

Strengths

- Three-annotator majority-vote QC with two-round review [\[Q42, Q45, Q109\]](#)
- Post-review Fleiss Kappa 0.912–1.00 across all languages [\[Q48\]](#)
- Detailed annotation guide with explicit disambiguation rules [\[Q149, Q152, Q153, Q154, Q155\]](#)
- Native-speaker annotators across all languages [\[Q7\]](#)
- Per-country remuneration via professional logistics company [\[Q44\]](#)

ID	Evidence
Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
Q149	"This section provides the complete annotation guide and instruction for annotators working for labeling all slots types." (p.20)
Q152	"When annotating Bank names, you do not need to include "bank" unless it is attached to the bank name, like seen above, with Ecobank." (p.21)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
Q155	"You include "bill" as part of the annotation." (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground truth was set by pan-African native-speaker crowdworkers, not by EEU/AAWSA customer service agents as the deployment requires. The deployment elicitation explicitly states utility agent/billing staff judgment determines correctness.

OC-2: Likely substantial for utility-specific spans (BILL_TYPE boundaries, place-name resolution to kebele/sub-city). D7 ('የቦሎ ማሳደሻ' annotated as BILL_TYPE) — whether this matches EEU/AAWSA categorization cannot be confirmed without domain-expert review. Cultural conventions around formal Amharic register also vary by service institution.

OC-3: Demographics partially documented: native speakers via Kenya-based logistics company [\[Q44\]](#); per-language counts and Ethiopia-specific recruitment unspecified (NOT FOUND in YAML benchmark_amharic_split_details). No domain-expertise requirement documented.

OC-4: Re-annotation by EEU and AAWSA staff (or supervised review) is needed for any utility-specific slot labels before deployment use. The deployment explicitly requires this.

OC-5: Aggregation method is majority vote with review [\[Q45\]](#); minority-perspective erasure risk is moderate, particularly for ambiguous slot types where annotators disagreed [\[Q53\]](#). Reviewer identity and resolution rules are not detailed.

OC-6: Convergent validity concern: Amharic ground-truth labels may not correlate with EEU/AAWSA agent judgments. External validity concern: under-annotation observed in samples [\[D15, D16\]](#) reduces reliability of slot-filling F1 as a deployment-readiness signal.

ID	Evidence
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Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q53	"Examination of annotator feedback and comparative analysis between unreviewed and reviewed versions indicated that ambiguous slot types significantly impacted annotation quality and introduced unnecessary complexity." (p.4)
D7	"የቦሌ ማሳደሻ ስንት ነው መክፈል የሚኖርብኝ?" — BILL_TYPE: የቦሌ ማሳደሻ — "How much do I need to pay for Bole road expansion?" — bill query referencing Bole (Addis Ababa neighborhood)
D15	"ነገ ጠዋት 12 ሰዓት የቀጠርኩትን ማንቁያ በደንብ እንዲሰማኝ ድምጽ ጨምርሰኝ" — spans: [] — "Increase the volume of the alarm I set for 12 o'clock tomorrow morning so I can hear it well." — TIME and DATE slots present in text but no spans annotated
D16	"የእናቴን እና የእባቴን የልደት ቀን የቀን መቁጠሪያዬ ላይ ስላቀያየርኩት አስተካክለው" — spans: [] — "Fix it because I changed my mother's and father's birthday on my calendar." — CALENDAR_EVENT-like content but no spans annotated
WEB-9	INJONGO paper — gap_id 3 NOT FOUND on Amharic annotator domain expertise; gap_id 7 NOT FOUND on Amharic-specific Kappa (https://arxiv.org/abs/2502.09814)

Information Gaps

- Amharic-specific pre- and post-review Fleiss Kappa values (gap_id 7)
- Per-language annotator counts and locations for Amharic (NOT FOUND)
- Whether reviewers had domain expertise in any specific industry

Requires Expert Verification

- Whether EEU/AAWSA agents agree with sampled BILL_TYPE boundary decisions in Amharic data
- Whether under-annotated examples [D15, D16] are isolated or systematic across the full Amharic split

Remediation

Gap 1: Annotators were pan-African native-speaker crowdworkers with no documented utility domain expertise; deployment requires EEU/AAWSA agent validation as ground-truth authority.

Recommendation: Have EEU and AAWSA customer service agents re-validate a stratified sample of the Amharic test split's utility-relevant labels; resolve discrepancies; retain only utility-domain examples whose labels match agent judgment for deployment-readiness evaluation.

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

LOWER-priority dimension with strong alignment. Both benchmark and deployment use structured text output (intent label + slot extraction) with standard text representation. Intent detection is scored by accuracy and slot-filling by F1 [Q84], and the structured ``-separated extraction format [Q146] is a reasonable text representation that maps to programmatic parsing. Models are instructed to return `` only when no entities are found [Q145], handling the empty-output case explicitly. Five seeds for fine-tuning [Q63] and five prompt templates for LLM eval [Q73] provide robustness. The output modality is text-only, matching deployment exactly. Minor concerns: the `` separator is benchmark-specific syntax that requires post-processing in production, and the text-string output form can complicate confidence-score extraction useful for chatbot dialogue management. F1 metric is appropriate for span extraction.

ID	Evidence
Q63	"All results are averaged over five seeds." (p.5)
Q73	"The evaluations of LLMs use 5 different prompting templates and a temperature of 0.5." (p.6)
Q84	"Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages." (p.7)
Q145	"Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return `` only." (p.19)
Q146	"Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater" (p.19)

Strengths

- Intent accuracy and slot F1 are standard, well-understood metrics [Q84]
- Five-seed averaging for fine-tuning and five-template averaging for LLMs reduce evaluation variance [Q63, Q73]
- Structured output format with explicit empty-case handling [Q145, Q146]
- Text-only modality matches Telegram/SMS deployment exactly
- Both fine-tuning and prompting paradigms evaluated [Q61, Q65], allowing the user to choose deployment-appropriate models

ID	Evidence
Q61	"We evaluate three categories of models: (1) encoder-only models such as XLM-RoBERTa Large (Conneau et al., 2019), AfroXLMR (Alabi et al., 2022), AfroXLMR-76L (Adelani et al., 2023a), AfriBERTa V2 (Oladipo, 2024), (2) encoder-decoder models such as mT5-Large (Xue et al., 2020), AfriTeVa V2 Large (Oladipo et al., 2023), and (3) a multilingual variant of LLM2Vec model (BehnamGhader et al., 2024) i.e. NLLB-LLM2Vec (Schmidt et al., 2024) that stack NLLB-encoder (NLLB Team et al., 2022) model with LLaMa 3.1 8B (Grattafiori et al., 2024) to develop a multilingual sentence transformer model." (p.5)
Q63	"All results are averaged over five seeds." (p.5)
Q65	"First, we conduct zero-shot prompting using the following widely used LLMs for evaluation: GPT4o, Gemini 1.5 Pro (Reid et al., 2024), Gemma 2 9B/27B Instruct (Team et al., 2024), Llama 3.1 8B/3.3 70B Instruct (Grattafiori et al., 2024), and Aya-101 (Üstün et al., 2024)." (p.5)
Q73	"The evaluations of LLMs use 5 different prompting templates and a temperature of 0.5." (p.6)

Q84	“Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages.” (p.7)
Q145	“Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return `\$\$` only.” (p.19)
Q146	“Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater” (p.19)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text output for intent + slots matches Telegram/SMS chatbot needs. No speech, image, or open-ended generation required.

OF-2: N/A — TTS not required; deployment is text-based.

OF-3: Adult literacy in Addis Ababa is high (male ~93.6%, female ~80% per 2005 baseline [\[WEB-2, WEB-4\]](#)) — text-based interaction is feasible for the target population. Telegram dominance and 4G coverage >80% [\[WEB-6\]](#) support text channel.

OF-4: Minor: `\$\$`-separated string output [\[Q146\]](#) requires production-side parsing logic; no probabilistic confidence scores in benchmark output schema, which is sub-optimal for dialogue-management uncertainty handling but does not invalidate evaluation.

ID	Evidence
Q84	“Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages.” (p.7)
Q145	“Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return `\$\$` only.” (p.19)
Q146	“Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater” (p.19)
WEB-2	Addis Ababa adult literacy >93% male, ~80% female — text-based interaction feasible (https://worldpopulationreview.com/cities/ethiopia/addis-ababa)
WEB-4	Wikipedia Demographics of Addis Ababa corroborates literacy figures (https://en.wikipedia.org/wiki/Addis_Ababa)
WEB-6	4G coverage >80% in primary deployment cities — text channel infrastructure available (https://www.mordorintelligence.com/industry-reports/ethiopia-telecom-mno-market)

Information Gaps

- Whether F1 with macro vs micro averaging is reported by slot type or language
- Whether confidence scores can be extracted from fine-tuned models for production uncertainty handling